

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 02: Friday 14 August 2026

(Week 1)

Remark on Assessment: Unlike the midterm test score, there will not be any adjustment to the final exam score.



Course Schedule:

Week 1 (Aug 10 - Aug 14) —

Review of 1-variable Calculus

Read: Thomas' Calculus 15th ed. chapters 1-9

Review: MIT course 18.01 Single Variable Calculus,
taught in Fall 2006 by Prof. David Jerison

Topics to be reviewed in Week 1:

1. Functions, limits and continuity (Chapter 1-2 of textbook (Thomas' Calculus))
2. Derivatives / Differentiation (Chapter 3)
3. Application of Derivatives (Chapter 4)
4. Integrals (Chapters 5, 7, 8)
5. Applications of Definite Integrals (Chapter 6)
6. Power series, Maclaurin Series, Taylor Series (Chapter 9) (only if we have time this week.)

2. Derivatives / Differentiation (Chapter 3)

Q01.9. Let $f(x) = x^2$. Find $f'(x)$.

Q01.10. Let $f(x) = x^3 + x^2 \tan x$. Find $f'(x)$.

Q01.11. Differentiate $\frac{x^2}{\ln x}$ with respect to x .

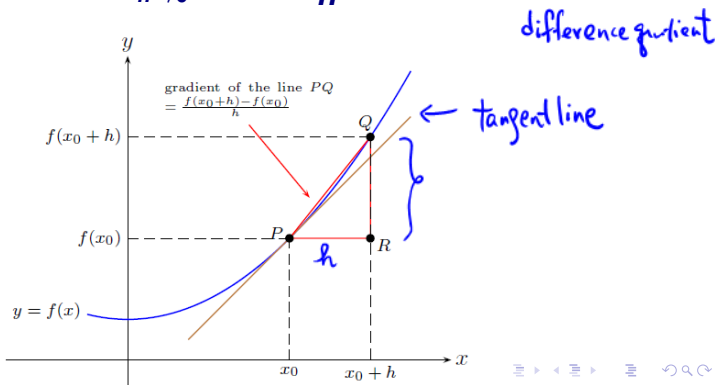
Q01.12. Let $y = \tan(t^3 + 2t)$. Find $\frac{dy}{dt}$.

2. Derivatives / Differentiation

(Thomas' Calculus, Chapter 3)

Recall: The **derivative** of a function f at the point x_0 , denoted by $f'(x_0)$, is given by the following limit

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$



When $f'(x_0)$ exists, we say that f is differentiable at $x = x_0$.

Geometrically in this case, $f'(x_0)$ is equal to the slope of the tangent line to the curve $y = f(x)$ at the point $(x_0, f(x_0))$.

Remark. Alternatively, by writing $h = \Delta x$, one has

$\Delta x = \text{change in } x$
 $\Delta y = \text{change in } y \text{ or } f(x)$

$$\begin{aligned} f'(x_0) &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}. \end{aligned}$$

Remark. Thus $f'(x_0)$ gives the rate of change of the function f with respect to the variable x at $x = x_0$. (See Section 3.4)

Suppose $f'(x)$ exists for all x in an open interval I . We can then treat $f'(x)$ as a function defined on I .

The process of finding the derivative of a function is called **differentiation**.

If $y = f(x)$, we also write $\frac{d}{dx}f(x)$, $\frac{dy}{dx}$ or $\frac{df}{dx}$ to denote $f'(x)$.

In summary, we have

$$\frac{d}{dx}f(x) = \frac{dy}{dx} = \frac{df}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Q01.9. Let $f(x) = x^2$. Find $f'(x)$.

Solution:

$$\begin{aligned} f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^2 - x^2}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cancel{x^2} + 2x \cdot \cancel{\Delta x} + (\Delta x)^2 - \cancel{x^2}}{\cancel{\Delta x}} \\ &= \lim_{\Delta x \rightarrow 0} (2x + \Delta x) = 2x + 0 \\ &= 2x. \end{aligned}$$

$f'(3) = 2 \cdot 3 = 6$

Some standard derivatives

(See also the table (Differentiation Rules) near 2nd last page of Thomas' Calculus)

Function	Derivative
x^n	nx^{n-1}
$\cos(x)$	$-\sin(x)$
$\sin(x)$	$\cos(x)$
$\tan(x)$	$\sec^2(x)$
$\sec(x)$	$\sec(x) \tan(x)$

Function	Derivative
$\csc(x)$	$-\csc(x)\cot(x)$
$\cot(x)$	$-\csc^2(x)$
e^x	e^x
$\ln(x)$	$\frac{1}{x}$
$\sin^{-1}(x)$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}(x)$	$-\frac{1}{\sqrt{1-x^2}}$

Function	Derivative
$\tan^{-1}(x)$	$\frac{1}{1+x^2}$
$\cot^{-1}(x)$	$-\frac{1}{1+x^2}$
$\sec^{-1}(x)$	$\frac{1}{ x \sqrt{x^2-1}}, x > 1$
$\csc^{-1}(x)$	$-\frac{1}{ x \sqrt{x^2-1}}, x > 1$

Rules of Differentiation (Section 3.3)

Let u and v be differentiable functions of x , and let c be a constant.

Constant Rule	$\frac{d}{dx}(c) = 0$
Constant Multiple Rule	$\frac{d}{dx}(cu) = c \frac{du}{dx}$
Sum Rule	$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$
Product Rule	$\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$
Quotient Rule	$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\frac{du}{dx}v - u\frac{dv}{dx}}{v^2}$

$$y = f(g(x))$$

Chain Rule (Section 3.4)

Suppose $y = \underline{f(u)}$ is a differentiable function of u , and $\underline{u = g(x)}$ is a differentiable function of x . Then we have

Chain Rule	$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
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with $\frac{dy}{du}$ evaluated at $u = g(x)$.

The Chain Rule can also be written directly as

$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x).$

Q01.10. Let $f(x) = x^3 + x^2 \sin x$. Find $f'(x)$.

Solution: By the sum rule and product rule,

$$\begin{aligned}& \frac{d}{dx}(x^3 + x^2 \sin x) \\&= \frac{d}{dx}x^3 + \frac{d}{dx}(x^2 \sin x) \\&= 3x^2 + \frac{d}{dx}(x^2) \cdot \sin x + x^2 \cdot \frac{d}{dx} \sin x \\&= 3x^2 + 2x \sin x + x^2 \cos x.\end{aligned}$$

Q01.11. Differentiate $\frac{x^2}{\ln x}$ with respect to x .

Solution. By the quotient rule,

$$\begin{aligned}\frac{d}{dx} \left(\frac{x^2}{\ln x} \right) &= \frac{\frac{d}{dx}(x^2) \cdot \ln x - x^2 \frac{d}{dx}(\ln x)}{(\ln x)^2} \\&= \frac{2x \cdot \ln x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} \\&= \frac{2x \ln x - x}{(\ln x)^2}.\end{aligned}$$

Q01.12. Let $y = \tan(t^3 + 2t)$. Find $\frac{dy}{dt}$.

Solution: Let $y = \tan u$, $u = t^3 + 2t$.

$$\begin{aligned}\frac{dy}{dt} &= \frac{dy}{du} \cdot \frac{du}{dt} \\ &= \sec^2 u \cdot (3t^2 + 2) \\ &= \left(\sec^2(t^3 + 2t) \right) \cdot (3t^2 + 2)\end{aligned}$$

Q02.1. Consider the curve $2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$. Find

$\frac{dy}{dx}$ (in terms of x and y).

Think of $y = f(x)$

$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$
coordinate plane.

Q02.2. Derive the formula $\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$.

Q02.3. Let $f(x) = \tan^{-1}(x)$. Find $f''(x)$.

Q02.4. Recall from **Q02.1** the curve $C : 2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$.

(a) Find an equation of the tangent line to the curve C at the point $(1, 2)$.

(b) Find an equation of the normal line to the curve C at the point $(1, 2)$.

Q02.1. Consider the curve $2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$. Find $\frac{dy}{dx}$ (in terms of x and y). $y = f(x)$.

Solution: Differentiating both sides of the equation w.r.t. x and using the chain rule, we have

Implicit differentiation

$$\begin{aligned}\frac{d}{dx}(y^3) &= \frac{d}{dy}(y^3) \cdot \frac{dy}{dx} \\ &= 3y^2 \cdot \frac{dy}{dx}\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(2x^3 + y^3) &= \frac{d}{dx}(5xy) \\ \Rightarrow 6x^2 + 3y^2 \frac{dy}{dx} &= 5y + 5x \frac{dy}{dx} \\ \Rightarrow (3y^2 - 5x) \cdot \frac{dy}{dx} &= 5y - 6x^2 \\ \Rightarrow \frac{dy}{dx} &= \frac{5y - 6x^2}{3y^2 - 5x}.\end{aligned}$$

Q02.2. Derive the formula $\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$.

Solution: Let $y = \tan^{-1}(x)$. Then

$$\tan y = x. \quad \frac{d}{dx}(\tan y)$$

Differentiating both sides w.r.t. x , we have $= \frac{d}{dy}(\tan y) \cdot \frac{dy}{dx}$

$$\begin{aligned} \frac{d}{dx}(\tan y) &= \frac{d}{dx}(x) \\ \Rightarrow \frac{d}{dy}(\tan y) \cdot \frac{dy}{dx} &= \frac{d}{dx}(x) \\ &= \sec^2 y \cdot \frac{dy}{dx} \end{aligned}$$

$$\Rightarrow (\sec^2 y) \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

Recall: $\sec^2 \theta = 1 + \tan^2 \theta$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}.$$

Q02.3. Let $f(x) = \tan^{-1}(x)$. Find $f''(x)$.

Solution: From **Q02.2**, we have

$$f'(x) = \frac{1}{1+x^2} \cdot \frac{d}{dx} \left(\frac{1}{1+x^2} \right) = \frac{0 \cdot (1+x^2) - 1 \cdot (2x)}{(1+x^2)^2}$$

Then

$$\begin{aligned} f''(x) &= \frac{d}{dx}(f'(x)) \\ &= \frac{d}{dx} \left(\frac{1}{1+x^2} \right) = \frac{d}{dx} \left((1+x^2)^{-1} \right) \\ &= -\frac{1}{(1+x^2)^2} \cdot \frac{d}{dx}(1+x^2) \\ &= -\frac{2x}{(1+x^2)^2}. \end{aligned}$$

Notation. Let $y = f(x)$. Then the second derivative of $f(x)$ with respect to x is defined by

$$f''(x) = \frac{d}{dx} \left(\frac{d}{dx} (f(x)) \right),$$

and it may be denoted by:

$$f''(x) = f^{(2)}(x) = \frac{d^2}{dx^2} f(x) = \frac{d^2 y}{dx^2} = y'' = D^2 f(x).$$

In general, we can define the n -th order derivative of f for any positive integer n .

For example, the third derivative of $y = f(x)$ is defined by

$$f^{(3)}(x) = f'''(x) = \frac{d}{dx} f^{(2)}(x) = \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{d}{dx} (f(x)) \right) \right).$$

We shall denote the ***n***-th order derivative of ***f*** by

$$f^{(n)}(x) = \frac{d^n y}{dx^n} = D^n f(x).$$

Example 02.3. Let $f(x) = \tan^{-1}(x)$. Find $f^{(3)}(x)$.

Solution:

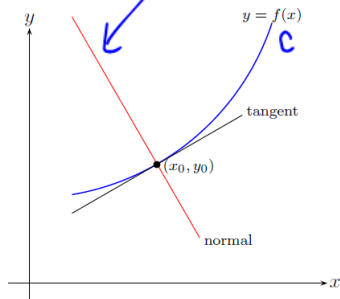
$$\begin{aligned} f^{(3)}(x) &= \frac{d}{dx}(f''(x)) \\ &= \frac{d}{dx}\left(-\frac{2x}{(1+x^2)^2}\right) \\ &= -\frac{\frac{d}{dx}(2x) \cdot (1+x^2)^2 - 2x \cdot \frac{d}{dx}((1+x^2)^2)}{((1+x^2)^2)^2} \\ &= -\frac{2 \cdot (1+x^2)^2 - 2x \cdot 2 \cdot (1+x^2) \cdot 2x}{(1+x^2)^4} \\ &= \frac{6x^2 - 2}{(1+x^2)^3}. \end{aligned}$$

3. Application of Derivatives (Chapter 4)

Q02.4. Recall from **Q02.1** the curve $C : 2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$.

(a) Find an equation of the tangent line to the curve C at the point $(1, 2)$.

(b) Find an equation of the normal line to the curve C at the point $(1, 2)$.



Solution: (a) Recall from **Q02.1** that $\frac{dy}{dx} = \frac{5y - 6x^2}{3y^2 - 5x}$.

At **(1, 2)**, we have

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{5 \cdot 2 - 6 \cdot 1^2}{3 \cdot 2^2 - 5 \cdot 1} = \frac{4}{7}.$$

Thus the slope of the tangent line to **C** at (1, 2) is $\frac{4}{7}$. $= m$

Thus an equation of the tangent line is:

$$\underline{y - 2 = \frac{4}{7}(x - 1)}.$$

(b) Find an equation of the normal line to the curve **C** at the point (1, 2).

Solution: Note that the tangent line and normal line to **C** at **(1, 2)** are perpendicular to each other, so that product of their slopes is -1 .

From (a), slope of the tangent line is $\frac{4}{7}$. Thus slope of the normal line is $-\frac{7}{4}$.

Thus an equation of the normal line is:

$$y - 2 = -\frac{7}{4}(x - 1).$$

$$\text{slope} = -\frac{7}{4}$$

(1, 2)

(linear approximation)

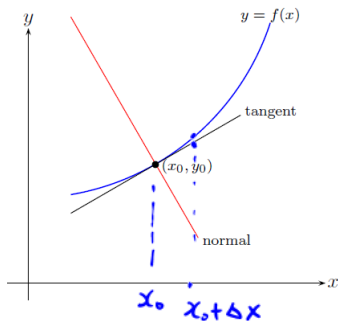
Q02.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(1) = 2$ and $f'(1) = 3$. Find an approximate value of $f(1.05)$.

Q02.6. (a) Give a sketch of the graph of a function $f(x)$ defined on an interval I such that $f'(x) < 0$ for all $x \in I$.

(b) Give a sketch of the graph of a function $g(x)$ defined on an interval I such that $g'(x) > 0$ for all $x \in I$.

Linear approximation

$$\text{Recall: } f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$



When Δx is small, $\Delta y \approx f'(x_0) \cdot \Delta x$.

$$f'(x_0) \approx \frac{\Delta y}{\Delta x}$$

$$f(x) - f(x_0) \approx f'(x_0) \cdot \Delta x$$

$$f(x) \approx \underbrace{f(x_0) + f'(x_0) \cdot \Delta x}$$

Q02.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(1) = 2$ and $f'(1) = 3$. Find an approximate value of $f(1.05)$.

Solution: $x_0 = 1$, $\Delta x = 1.05 - 1 = 0.05$

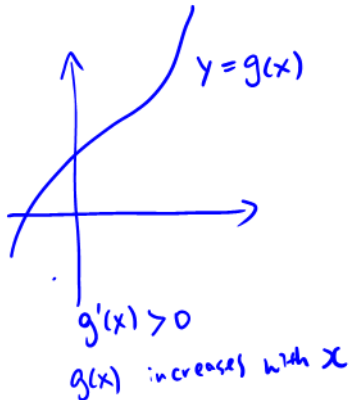
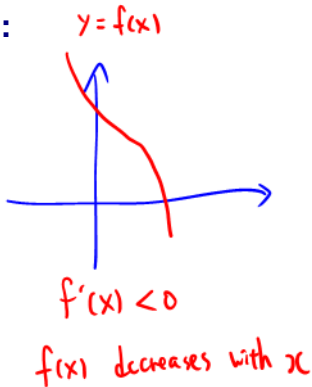
$$\begin{aligned} f(1.05) &\approx f(1) + f'(1) \cdot \Delta x \\ &\approx 2 + 3 \cdot 0.05 \\ &\approx 2.15. \end{aligned}$$

Q02.6. (a) Give a sketch of the graph of a function $f(x)$ defined on an interval I such that $f'(x) < 0$ for all $x \in I$.

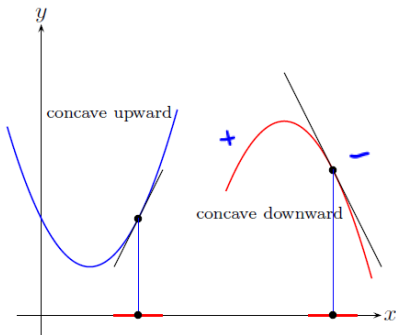
(b) Give a sketch of the graph of a function $g(x)$ defined on an interval I such that $g'(x) > 0$ for all $x \in I$.

Solution:

(a)



Q02.7. Suppose a function f is such that $f''(x) < 0$ for all x in an interval. Which of the following two curves can be the graph of $f(x)$? Blue curve or Red curve?



Solution:

$f'(x)$ increases
with x
 $\therefore f''(x) > 0$ on I

$f'(x)$ decreases with x
 $f''(x) < 0$ on I

Q02.8. Consider the function $f(x) = x^3 - 3x$.

(a) Find the critical points of $f(x)$.

(b) Find the local maximum point(s) and local minimum point(s) of $f(x)$. Find also the corresponding local maximum value(s) and local minimum value(s) of $f(x)$.

Q02.9. Consider the function

$$f(x) = x^2 e^x, \quad x \in [-3, 2].$$

(i) Find the critical points of $f(x)$ on the interval $[-3, 2]$.

(ii) Find the absolute maximum value of $f(x)$ on the interval $[-3, 2]$.

Q02.8. Consider the function $f(x) = \underline{x^3 - 3x}$.

(a) Find the critical points of $f(x)$.

(b) Find the local maximum point(s) and local minimum point(s) of $f(x)$. Find also the corresponding local maximum value(s) and local minimum value(s) of $f(x)$.

Solution: (a) $f(x) = x^3 - 3x$.

$$f'(x) = 3x^2 - 3$$

To get critical points of $f(x)$,

$$f'(x) = 0$$

$$\Leftrightarrow 3(x^2 - 1) = 0$$

$$\Leftrightarrow 3(x+1)(x-1) = 0 \quad \Leftrightarrow x = -1, 1.$$

$f(x)$ has critical points at $x = -1, 1$.

$$b) \quad f'(x) = 3x^2 - 3$$

$$f''(x) = 6x.$$

$$\text{At } x=1, \quad f''(1) = 6 \cdot 1 = 6 > 0$$

$f(x)$ has a loc. min. at $x=1$

$$f(1) = 1^3 - 3 = 2.$$

$$\text{At } x=-1, \quad f''(-1) = 6 \cdot (-1) = -6 < 0$$

$\therefore f(x)$ has a loc. max point at $x=-1$.

$$f(-1) = (-1)^3 - 3(-1) = 2.$$

Q02.9. Consider the function

$$f(x) = x^2 e^x, \quad x \in [-3, 2].$$

- (i) Find the critical points of $f(x)$ on the interval $[-3, 2]$.
(ii) Find the absolute maximum value of $f(x)$ on the interval $[-3, 2]$.

Solution: (i) $f'(x) = 2x \cdot e^x + x^2 \cdot e^x$
 $= e^x x(x+2).$

$f(x)$ has critical point at
 $f'(x) = 0$

$$\Leftrightarrow e^x x(x+2) = 0$$

$$\Leftrightarrow x = 0, -2 \quad (\because e^x > 0).$$

(ii) $f(x)$ attains abs. max value
either at a critical point or at an endpoint.

$$f(-3) = (-3)^2 e^{-3} \approx 0.45$$

$$f(-2) = (-2)^2 e^{-2} \approx 0.5$$

$$f(0) = 0$$

$$f(2) = 2^2 e^2 \approx 28 \dots$$

$f(x)$ attains abs. max value at $x=2$

$$\text{with } f(2) = 2^2 e^2 \approx 28 \dots$$

Q02.10. Find $\lim_{x \rightarrow 1} \frac{e^x - e}{x^2 - 1}$.

} L'Hopital's Rule

Q02.11. Find $\lim_{x \rightarrow \infty} \frac{x^3 + x}{e^x + x^2}$.

Q02.12. Find an antiderivative of the function $f(x) = \cos x$.

Q02.13. Find the following indefinite integrals:

(i) $\int \sin x \, dx$, (ii) $\int x^2 \, dx$, (iii) $\int (2x + 3)^4 \, dx$.

Q02.10. Find $\lim_{x \rightarrow 1} \frac{e^x - e}{x^2 - 1}$.

$$\frac{0}{0} \leftarrow \text{indeterminate}$$

$$\begin{aligned} \text{L'Hopital} &= \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(e^x - e)}{\frac{d}{dx}(x^2 - 1)} = \lim_{x \rightarrow 1} \frac{e^x}{2x} \\ &= \frac{e^1}{2} = \frac{e}{2} \end{aligned}$$

Q02.11. Find $\lim_{x \rightarrow \infty} \frac{x^3 + x}{e^x + x^2}$.

$$\frac{\infty}{\infty}$$

← indeterminate

L'Hopital $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{e^x + 2x}$

$$\frac{\infty}{\infty}$$

L'Hopital $\lim_{x \rightarrow \infty} \frac{6x}{e^x + 2}$

$$\frac{\infty}{\infty}$$

L'Hopital $\lim_{x \rightarrow \infty} \frac{6}{e^x}$

$$= 0.$$

~~$\lim_{x \rightarrow 1} \frac{x^2}{x}$
" $\lim_{x \rightarrow 1} \frac{2x}{1}$~~

4. Integrals (Chapters 5, 7, 8)

Indefinite Integrals

Given a function $f(x)$ defined on an interval I . A function $F(x)$ is called an antiderivative of f on I if

$$F'(x) = f(x) \text{ for all } x \text{ in } I.$$

It turns out that any two anti-derivatives of a function $f(x)$ over I differ by a constant.

Thus to get all anti-derivatives of $f(x)$, we just need to find one anti-derivative $F(x)$ of $f(x)$. Then any antiderivative of f on I is of the form $F(x) + C$ for some constant C .

This is expressed as:

$$\int f(x) dx = F(x) + C.$$

Q02.12. Find an antiderivative of the function
 $f(x) = \cos x$.

Solution: An anti-derivative of $\cos x$
is $\sin x$

$$\text{because } \frac{d}{dx}(\sin x) = \cos x$$

$$\int \cos x \, dx = \sin x + C$$

Q02.13. Find the following indefinite integrals:

$$(i) \int \sin x \, dx = -\cos x + C$$

$$(\because \frac{d}{dx}(-\cos x) = \sin x)$$

$$(ii) \int x^2 \, dx = \frac{x^3}{3} + C$$

$$(iii) \int (2x + 3)^4 \, dx = \frac{(2x+3)^5}{5 \cdot 2} + C$$